2 **(a)** **(i)** Define gravitational field strength.

.....

..... [1]

(ii) Hence, state with a reason, if gravitational field strength is a scalar or vector quantity.

.....

..... [1]

(iii) State Newton's law of gravitation and use your definition in **(a)(i)** to write down an expression for the gravitational field strength *g* at a distance *R* from a point mass *M*.

.....

.....

.....

.....

..... [2]

(b) The mass of planet Jupiter is 1.90×10^{27} kg and its radius is 7.14×10^{7} m. The radius of Jupiter's orbit around the Sun is 7.79×10^{11} m. The mass of the Sun is 1.99×10^{30} kg.

(i) Calculate the ratio

$$\frac{\text{gravitational field strength on the surface of Jupiter due to the Sun}}{\text{gravitational field strength on the surface of Jupiter due to the mass of Jupiter}}.$$

ratio = ... [2]

(ii) Hence explain if the gravitational field strength due to the Sun on the surface of Jupiter can be neglected.

.....

[1]

.....

(c) The Galilean moons are the largest moons of Jupiter which were first discovered by Galileo in January 1610. Some of the data of the 3 Galilean moons closest to Jupiter are given in Fig. 2.1.

Name of moon	Io	Europa	Ganymede
Average orbital radius / m	4.22×10^8	6.71×10^8	1.07×10^9

Fig. 2.1

(i) For moons revolving around a planet in circular orbits, determine that the orbital period T of a moon is given by

$$T^2 = KR^3$$

where R is the radius of the orbit and K is a constant. Explain your working clearly.

[3]

(ii) Hence, show that the orbital period of Io : Europa : Ganymede is 1 : 2 : 4 approximately.

[2]